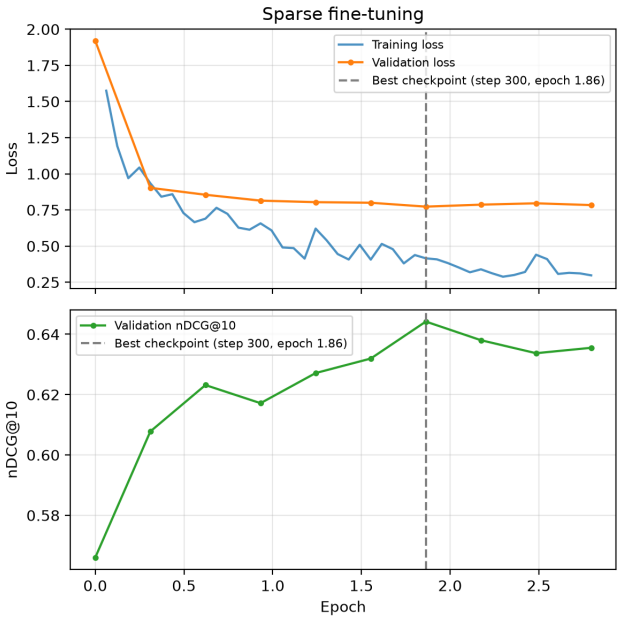
COS 487 / 587 Course Project Part 2: IR Rangers

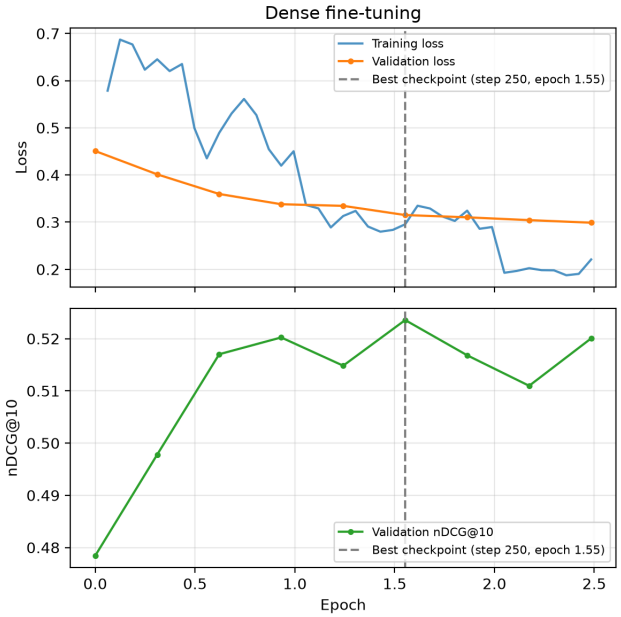
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(a) SPLADE.



(b) Dense (MiniLM).

Figure 1: Training and validation loss.

Abstract

Keywords

information retrieval, BM25, SPLADE, dense retrieval, rank fusion, scientific document retrieval, passage retrieval

1 Introduction
2 Methodology
2.1 Preprocessing
2.2 BM25 Baseline
2.3 Training Data
2.4 Learned Sparse Retrieval
2.5 Dense Retrieval
2.6 Paper Ranking
2.7 Rank Fusion

$$\mathrm{RRF}(d) = \sum_{m \in M} \frac{1}{k + \mathrm{rank}_m(d)}, \qquad k = 60, \tag{1}$$

2.8 Evaluation Setup
3 Evaluation Results
3.1 Training
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4.7 Strengths and Weaknesses
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Table 1: Effectiveness on the 250 Study queries. Strict metrics count only grade-2 documents (238 queries). Superscripts mark a significant improvement over the model with that letter (paired t-test, $p < 0.05$).

	Model	nDCG	nDCG@5	MRR	P@5	P@10	R@100	Strict MRR	Strict P@5
Papers									
a	BM25	0.582	0.514	0.544	0.137	0.080	0.803	0.535	0.124
b	Dense	0.640^{a}	0.592^{a}	0.631^{a}	0.156^{a}	0.084	0.807	0.606^{a}	0.141^{a}
c	SPLADE	$\mathbf{0.671}^{a}$	$\mathbf{0.631}^{ab}$	$\mathbf{0.658}^{a}$	$\mathbf{0.167}^{ab}$	$\mathbf{0.090}^{ab}$	**0.825**	$\mathbf{0.645}^{a}$	$\mathbf{0.151}^{ab}$
Paragraphs									
a	BM25	0.474	0.416	0.516	0.174	0.107	0.592	0.450	0.115
b	Dense	0.447	0.377	0.477	0.175	0.104	0.607	0.415	0.115
c	SPLADE	$\mathbf{0.494}^{b}$	$\mathbf{0.434}^{b}$	$\mathbf{0.539}^{b}$	**0.190**	$\mathbf{0.116}^{b}$	$\mathbf{0.630}^{a}$	$\mathbf{0.484}^{b}$	$\mathbf{0.135}^{ab}$

Table 2: Zero-shot versus fine-tuned neural models on Study queries. † marks a significant difference between the two versions of a model (paired t-test, $p < 0.05$).

	Papers					Paragraphs				
Model	nDCG	nDCG@5	MRR	P@5	R@100	nDCG	nDCG@5	MRR	P@5	R@100
Dense, zero-shot	0.635	0.591	0.619	$0.166^{\dagger}$	0.812	0.458	0.385	0.484	0.174	0.633
Dense, fine-tuned	0.640	0.592	0.631	0.156	0.807	0.447	0.377	0.477	0.175	0.607
SPLADE, zero-shot	0.668	0.625	0.654	0.161	0.815	0.485	0.431	0.524	0.186	0.603
SPLADE, fine-tuned	0.671	0.631	0.658	0.167	0.825	0.494	0.434	0.539	0.190	0.630

Table 3: Reciprocal rank fusion on Study queries. Superscripts mark a significant improvement over the run with that letter (paired t-test, $p < 0.05$).

	Run	nDCG	nDCG@5	MRR	P@5	P@10	R@100	Strict MRR	Strict P@5
Papers									
a	BM25	0.582	0.514	0.544	0.137	0.080	0.803	0.535	0.124
b	SPLADE	0.671^{a}	0.631^{a}	0.658^{a}	0.167^{a}	0.090^{a}	0.825	0.645^{a}	0.151^{a}
c	RRF(BM25+SPLADE)	0.675^{a}	0.628^{a}	0.659^{a}	0.170^{a}	0.094^{a}	0.834^{a}	0.646^{a}	0.149^{a}
d	RRF(all three)	$\mathbf{0.701}^{abc}$	$\mathbf{0.655}^{ac}$	$\mathbf{0.689}^{ac}$	$\mathbf{0.174}^{a}$	$\mathbf{0.095}^{ab}$	$\mathbf{0.853}^{abc}$	$\mathbf{0.673}^{ac}$	$\mathbf{0.154}^{a}$
Paragraphs									
a	BM25	0.474	0.416	0.516	0.174	0.107	0.592	0.450	0.115
b	SPLADE	0.494	0.434	0.539	0.190	0.116	0.630^{a}	0.484	0.135^{a}
c	RRF(BM25+SPLADE)	0.531^{ab}	0.468^{ab}	0.569^{a}	0.201^{a}	0.119^{a}	0.666^{ab}	0.508^{a}	0.136^{a}
d	RRF(all three)	$\mathbf{0.550}^{abc}$	$\mathbf{0.489}^{abc}$	$\mathbf{0.582}^{ab}$	$\mathbf{0.218}^{abc}$	$\mathbf{0.130}^{abc}$	$\mathbf{0.689}^{abc}$	$\mathbf{0.523}^{ab}$	$\mathbf{0.148}^{abc}$

Table 5: Our judgments of top-ranked Test paragraphs.

Grade	Dense	SPLADE	Both
Relevant (2)	TODO	TODO	TODO
Partially relevant (1)	TODO	TODO	TODO
Non-relevant (0)	TODO	TODO	TODO

Table 6: Index size, indexing time and per-query latency.

Model	Index size	Indexing	Query (ms)
BM25 (paragraphs)	TODO	TODO	TODO
SPLADE	TODO	TODO	TODO
Dense	TODO	TODO	TODO